


```

        %¼
[ Guided 6-Face Scan ] %Ä% % % % % % % % % % % % %
  (U -> R -> F -> D -> L -> B)          % (Rescan)
        %                               %
        %¼                               %
[ State Validation Engine ] % % (Invalid) % % (Error Modal Recovery)
        % (Valid)
        %¼
[ Sub-Second Kociemba Solve ]
        %
        %¼
[ Interactive 3D Playback & Step-by-Step Guidance ]

```

4. Key UI Components & Interactions

4.1 Guided Camera Scanner HUD (CameraScanner.tsx)

- Viewport Reticle: 3x3 rounded bounding grid centered over the live video stream.
- Stability Indicator: A circular progress ring around each sticker cell filling up over 10 consecutive stable frames.
- Color Confidence Bar: Real-time readout of CIELAB distance matching confidence.
- Face Progress Tracker: 6 segmented indicators at the top showing completed vs remaining faces.
- Orientation Guide Widget: Mini 3D preview cue showing how to rotate the physical cube to reach the next required face.

4.2 Mathematical Error Recovery Modal (ErrorRecoveryModal.tsx)

- Triggered automatically if state validation detects parity or count anomalies.
- Presents an unfolded 2D cube net view (U, L, F, R, B, D).
- Highlights problematic pieces with pulsing red outlines.
- Provides an intuitive color palette selector to manually override misclassified stickers.

4.3 Solve Playback Controls (SolveControls.tsx)

- Move Tape: Horizontal scrolling carousel of move tokens (e.g. [U] [R2] [F'] [D]). The current move is highlighted in bright cyan with enlarged typography.
- Playback Controls:
 - Play / Pause (toggleable with Spacebar).
 - Step Forward / Step Backward (Right Arrow / Left Arrow).
 - Speed Adjustment: 0.5x, 1x, 2x, 4x.
 - Reset to initial scramble (R).
 - Progress Gauge: Real-time percentage indicator and step counter (K / N).

4.4 Real-Time Computer Vision Debugger (CVDebugger.tsx)

- Collapsible diagnostics drawer for advanced users and developers.
- Displays live RGB, HSV, CIELAB color values for all 9 grid points.
- Frame rate monitor (FPS) and frame drop detection.

5. Responsive Design & Accessibility

- Desktop (>= 1024px): Dual-pane split view with video HUD/controls on the left and the 3D twin visualizer on the right.
- Mobile (< 1024px): Vertically stacked layout with sticky playback controls at the bottom viewport boundary.
- Accessibility:
 - All interactive buttons include descriptive aria-label attributes.
 - High contrast ratio exceeding WCAG AA standards (minimum 4.5:1 for normal text, 7:1 for large display metrics).
 - Keyboard navigation fully supported across all critical playback actions.

6. Phone Scanning & Review (ADR-010)

```

[ Desktop: Scan with Phone ] % % QR % % %° [ Phone: first-run coach (once) ]
        %                               %
        %¼                               %¼
[ Desktop: live 6-face progress ] %Ä% % photos % % [ Phone: guided capture, auto shutter ]
        % (all six solved)

```

```

    %¼
[ Desktop: review net ] % % Rescan face % % %° [ Phone jumps to that face ]
    % (Confirm & solve, valid cube only)
    %¼
[ Solve & animated playback ]           [ Phone: "Solving on your computer" ]

```

6.1 State Colors

One meaning per accent, everywhere in the flow:

Color	Meaning
Sky	Active, in progress, holding steady
Emerald	Verified: face read, cube valid
Amber	Needs attention: something to fix or check
Rose	Error: the cube cannot exist as entered

6.2 Phone Capture (/companion)

- First-run coach (CaptureCoach.tsx): three cards (hold, light, order). Shown once per device, reopenable from the help button.
- Step card: an isometric mini cube (FaceGuide.tsx) showing which center faces the camera, which is on top and which is on the right, plus a one-move instruction and an end-position check.
- Reticle: white idle, amber when something needs fixing, sky while holding steady, emerald once read. Live dots show that the phone sees each tile.
- Hint line: exactly one instruction, naming the fix rather than the fault ("Glare on the bottom row. Tilt the cube slightly.").
- Shutter: always usable by hand. A ring fills while every check holds; at 600 ms the photo takes itself. Auto-capture can be turned off.
- Layout: a column of header, guidance, reticle, hint and controls. The reticle is sized to the remaining height so nothing overlaps on short screens. Safe-area insets are respected and touch targets are at least 44 px.

6.3 Review (ScanReview.tsx)

- An unfolded net in the layout of a real cube, with amber rings on tiles that were hard to read.
- Selecting a tile opens an editor with that tile cropped from the user's own photo, six swatches (keys 1–6), and "Rescan this face".
- A live count of each color, and one sentence describing what to fix when the cube cannot exist.
- The screen says so when a photo was turned to fit, or when two photos show the same face.

6.4 Motion

Entrances take 150–320 ms with an ease-out curve (cb-rise, cb-turn-in, cb-flash in globals.css), and only one thing moves at a time. All of them, and the spinners, stop under prefers-reduced-motion.